Technical report: teacher-guided corrective residuals in BrainCAD

Andrés de Vicente, Renee Vieira, and Charlie Hou

Neuromatch Impact Scholars, Team RNNematode, 2026

1 Project overview

The original goal was to implement a sensory-cerebellar NCAP-like experiment in the BrainCAD locomotion stack. The main result became more specific: adding a biologically inspired module was not enough by itself. The learning signal determined whether the module became a real correction or another weak policy head. PPO-only spinal and residual circuits did not reliably learn rapid recovery. Teacher-guided residual imitation did. The method is intentionally close to residual imitation / policy distillation; the scientific point is that this structured teacher signal was necessary for the biologically inspired corrective architecture to become useful.

2 ReflexBench perturbation protocol

ReflexBench perturbs the agent during evaluation with three families of events:

$$F_t = s_F F_0 \mathbf{1}[t_0 \le t < t_0 + D_F] \qquad \text{external push,} \tag{1}$$

$$\mu_t = \mu_0(1 - s_\mu) \qquad \text{slip / lower friction,} \tag{2}$$

$$\tilde{o}_t = M_t \odot o_t + \epsilon_t \qquad \text{sensor corruption.} \tag{3}$$

Here s_F and s_μ are severity values, D_F is push duration, M_t is a dropout mask, and ϵ_t is noise. We verified that the perturbations were real by logging applied force magnitude, friction scale, and clean-versus-corrupted observation differences. Control episodes had zero event rates and zero observation corruption.

3 Policies and learning objectives

3.1 Cortex and teacher

The cortex is a stochastic actor-critic policy trained by PPO:

$$\max_\theta \; \mathbb{E}_t \left[\min(r_t(\theta)\hat{A}_t, \mathrm{clip}(r_t(\theta), 1-\epsilon, 1+\epsilon)\hat{A}_t) \right],$$

with a value loss and entropy term as in standard PPO [1]. The teacher has the same basic role as a cortex policy but is trained under domain-randomized perturbations. It is not used at test time in the residual student, except when evaluating the teacher as a baseline.

3.2 Cerebellar residual

The cerebellar module is bounded:

$$\Delta u_\psi(o_t, u_t^{ctx}) = \Delta u_{max} \tanh f_\psi([o_t, u_t^{ctx}]).$$

The optional gate is

$$\alpha_t = \sigma(g_\eta(o_t, u_t^{ctx}, \|\Delta u_t\|)).$$

The executed action is

$$u_t = \mathrm{clip}(u_t^{ctx} + \alpha_t \Delta u_t, u_{min}, u_{max}).$$

The supervised residual target is:

$$\Delta u_t^* = u_t^{teacher} - u_t^{ctx}.$$

For event and recovery masks m_t, training minimizes:

$$\mathcal{L}_{cb} = \frac{\sum_t m_t \|\Delta u_\psi(o_t, u_t^{ctx}) - \Delta u_t^*\|_2^2}{\sum_t m_t + \epsilon} + \lambda \|\psi\|_2^2.$$

This is the central difference between the successful model and the RL-only residual: the residual sees a dense vector target at exactly the times when recovery matters.

3.3 Structural consolidation

Consolidation trains a new cortex to imitate the final residual action:

$$u_t^{target} = \mathrm{clip}(u_t^{ctx} + \alpha_t \Delta u_t, u_{min}, u_{max}),$$

$$\mathcal{L}_{cons} = \frac{1}{N} \sum_t w_t \|\pi_{\phi'}(o_t) - u_t^{target}\|_2^2.$$

The consolidated policy is cortex-only at inference. It tests whether the correction can become part of the feedforward motor command, instead of requiring an extra residual module forever. This consolidation stage is also a useful guard against over-interpreting the residual module itself: if the correction can be folded into cortex, then the important ingredient is the training signal and target structure, not permanent extra inference-time machinery.

4 Statistics and sign conventions

For each perturbation family, we evaluated a grid of severities and summarized the curve by area under the curve (AUC). In the default Humanoid predefined severity sweep, push magnitudes ranged from 0 to 1 and were crossed with push duration, slip severity ranged from 0 to 1.25 and was crossed with patch width/duration, and sensor dropout bursts ranged from 0 to 0.4. For return,

$$\mathrm{AUC}_R = \int R(s)\, ds,$$

so higher values are better. For falls,

$$\mathrm{AUC}_F = \int P(\text{fall} \mid s)\, ds,$$

so lower values are better. The causal residual effect is always a paired comparison:

$$\Delta_F = \mathrm{AUC}_F(\mathrm{ON}) - \mathrm{AUC}_F(\mathrm{OFF}),$$

using the same checkpoint and the same perturbation schedule. A negative Δ_F therefore means the residual prevented falls. The same paired definition was used for return AUC. Confidence intervals in the saved tables are bootstrap intervals over seeds unless otherwise noted.

5 Main Humanoid results

Table 1 summarizes the N=20 Humanoid campaign. Negative fall deltas mean fewer falls under the same severity sweep. The important point is that the fall-AUC confidence interval excludes zero for push, slip, and sensor corruption, not only that the average looks favorable.

comparison	sweep_type	metric	mean	sem	n_seeds	ci
consolidated_minus_baseline	push	fall	-0.311	0.091	20	[-0.488, -0.146]
consolidated_minus_baseline	sensor	fall	-0.406	0.053	20	[-0.509, -0.306]
consolidated_minus_baseline	slip	fall	-0.316	0.090	20	[-0.496, -0.149]
consolidated_minus_baseline	push	return	59.218	16.307	20	[27.470, 88.720]
consolidated_minus_baseline	sensor	return	86.354	10.848	20	[65.226, 106.575]
consolidated_minus_baseline	slip	return	61.232	16.644	20	[26.948, 91.766]
imitation_on_minus_off	push	fall	-0.313	0.091	20	[-0.491, -0.147]
imitation_on_minus_off	sensor	fall	-0.447	0.056	20	[-0.553, -0.338]
imitation_on_minus_off	slip	fall	-0.325	0.092	20	[-0.511, -0.155]
imitation_on_minus_off	push	return	58.762	16.104	20	[27.600, 88.445]
imitation_on_minus_off	sensor	return	103.610	10.517	20	[82.712, 123.475]
imitation_on_minus_off	slip	return	61.357	16.605	20	[27.364, 91.611]

Table 1: Humanoid default predefined severity sweep, N=20.

6 Negative result: RL-only corrective subcircuits

The failure ladder matters because it rules out a simple architectural explanation. The useful ingredient was not the name of the module. SC-NCAP gain gating, PPO-trained residuals, curriculum residuals, BG-like gates, and stability auxiliary variants did not produce consistent causal robustness improvements. These results suggest that sparse reward was too indirect for learning a small, fast corrective residual. Conversely, the successful method should not be presented as a new distillation algorithm; it is a biologically organized use of a known supervised residual-learning idea.

The failed variants also differed mechanistically. In SC-NCAP, prediction error modulated sensory gain before the spinal interface. In the RL-only residual, a bounded torque correction was available but had to be discovered from sparse return. In the stability-auxiliary variants, the residual received extra pressure toward uprightness during perturbation windows, but the correction still did not become consistently useful. These attempts became useful controls: they show that the successful effect is not a generic consequence of adding more modules.

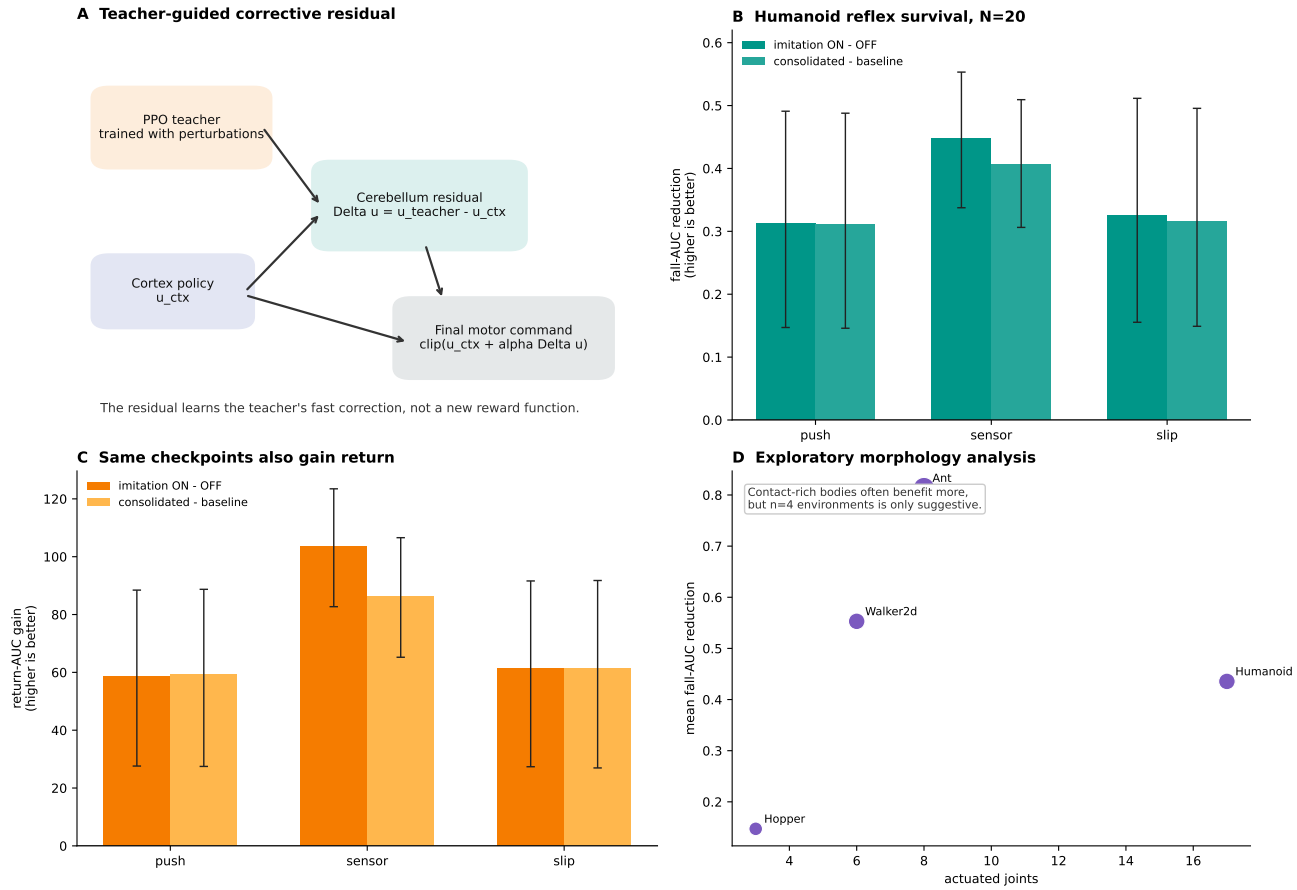

Figure 1: Main micropublication figure.

7 Perturbation sanity

We also checked that ReflexBench was not only logging intended perturbations. Table 2 reports simple on/off contrasts from the perturbation sanity runs. Push values are applied force differences, slip values are the drop in friction scale, and sensor values are clean-versus-corrupted observation differences, each summarized in the native scale of the environment and then reported compactly. Control episodes had zero event flags.

env	push	sensor	slip
humanoid	0.588	462.616	0.607
walker2d	0.494	11.713	0.538
ant	1.098	12.697	0.645
hopper	0.100	1.664	0.410

Table 2: Compact perturbation sanity checks. Nonzero push/slip/sensor contrasts confirm that the perturbations were actually applied.

8 Teacher quality and safety

Teacher degradation provided a useful counterfactual. Students trained from the full teacher improved fall robustness, while students trained from weak teacher checkpoints could become harmful. In the weak-teacher runs, larger residual corrections tended to predict worse stability outcomes in several windows. Thus, the residual pathway can inherit useful corrections, but it can also inherit a teacher's mistakes.

teacher_variant	sweep_type	n_seeds	fall_delta_ci	return_delta
T_full	push	3	-0.261 [-0.385, -0.152]	54.1
T_weak	push	3	0.122 [0.000, 0.281]	-20.5
T_full	sensor	3	-0.438 [-0.562, -0.375]	64.2
T_weak	sensor	3	0.104 [0.062, 0.125]	-13.1
T_full	slip	3	-0.608 [-1.000, -0.300]	111.6
T_weak	slip	3	0.366 [0.000, 0.622]	-37.8

Table 3: Teacher-quality replication. Negative fall deltas are helpful; positive fall deltas are harmful.

This threshold-like pattern helps interpret the result. If every teacher produced the same student benefit, the result would look like a generic capacity increase. Instead, the sign of the learned correction depends on teacher competence. This is consistent with the residual learning a corrective vector field rather than simply smoothing the action.

9 Structural consolidation

The consolidation result asks whether the residual is only useful as an extra online module. We collected trajectories from the residual-ON student and trained a new cortex policy to match the actually executed action. The consolidation target is:

$$u_t^{target} = u_t^{final}.$$

This is stricter than copying the teacher, because u_t^{final} includes the clipped combination of cortex, residual, and optional gate. In Humanoid, the consolidated cortex-only policy closed most of the baseline-to-imitation fall gap on the default schedule and also generalized under the long-push schedule. This gives the residual a training role: the correction first appears as a separate pathway, then can be folded back into the main motor command.

10 Cross-environment and morphology-dependent control demands

The morphology analysis should be treated cautiously. Hopper, Walker2d, Ant, and Humanoid are not an evolutionary tree, and four simulated bodies cannot establish a scaling law. Still, the pattern is useful as a control-demand analysis. More contact-rich bodies create more transient failure modes; this is exactly where layered corrective control should be useful. Ant and Humanoid show strong fall reductions. Hopper is a boundary condition: its underactuated morphology makes conservative corrections able to reduce some falls while hurting return.

env	source	metric	mean	sem	n
Ant-v4	consolidated_minus_baseline	fall	-0.778	0.045	3
Ant-v4	consolidated_minus_baseline	return	-3.953	1.119	3
Ant-v4	imitation_on_minus_off	fall	-0.814	0.024	3
Ant-v4	imitation_on_minus_off	return	-4.473	1.057	3
Hopper-v4	consolidated_minus_baseline	fall	-0.133	0.067	3
Hopper-v4	consolidated_minus_baseline	return	-15.659	0.263	3
Hopper-v4	imitation_on_minus_off	fall	-0.147	0.066	3
Hopper-v4	imitation_on_minus_off	return	-15.593	0.215	3
Walker2d-v4	consolidated_minus_baseline	fall	-0.553	0.064	3
Walker2d-v4	consolidated_minus_baseline	return	254.162	1.406	3
Walker2d-v4	imitation_on_minus_off	fall	-0.553	0.064	3
Walker2d-v4	imitation_on_minus_off	return	252.326	0.995	3
Humanoid-v4	imitation_on_minus_off	fall	-0.361	0.080	20
Humanoid-v4	imitation_on_minus_off	return	74.576	14.409	20
Humanoid-v4	consolidated_minus_baseline	fall	-0.344	0.078	20
Humanoid-v4	consolidated_minus_baseline	return	68.935	14.600	20

Table 4: Cross-environment summary. Humanoid rows use the N=20 default-schedule result; other environments use the saved cross-environment replication summaries.

env	joints	dof	contacts	imit fall delta	cons fall delta
Hopper-v4	3	6	1	-0.147	-0.133
Walker2d-v4	6	9	2	-0.553	-0.553
Ant-v4	8	14	4	-0.814	-0.778
Humanoid-v4	17	23	2	-0.435	-0.295

Table 5: Morphology table used for the exploratory scaling plot. Values may differ from Table 1 because this plot uses the cross-environment matched summary rather than the full Humanoid N=20 default-schedule summary. The regression confidence intervals are wide; this table is best read as motivation for future work rather than a final scaling law.

11 Additional checks

The report includes the most important controls in the main text: RL-only failures, perturbation sanity checks, teacher-quality degradation, and cross-environment summaries. The figure below collects these controls in one place so the direction of each check can be inspected visually.

12 Limitations

This project does not prove that the biological cerebellum implements our exact residual loss. The better claim is that a cerebellar-inspired supervised correction pathway is a useful engineering abstraction for these locomotion tasks. Modern privileged-adaptation baselines such as RMA remain important comparators. Compute fairness is also subtle: the teacher has its own perturbation-trained experience, while the student learns from that experience in supervised form. We therefore frame the result as a learning-signal result rather than a claim that this architecture dominates all adaptation methods.

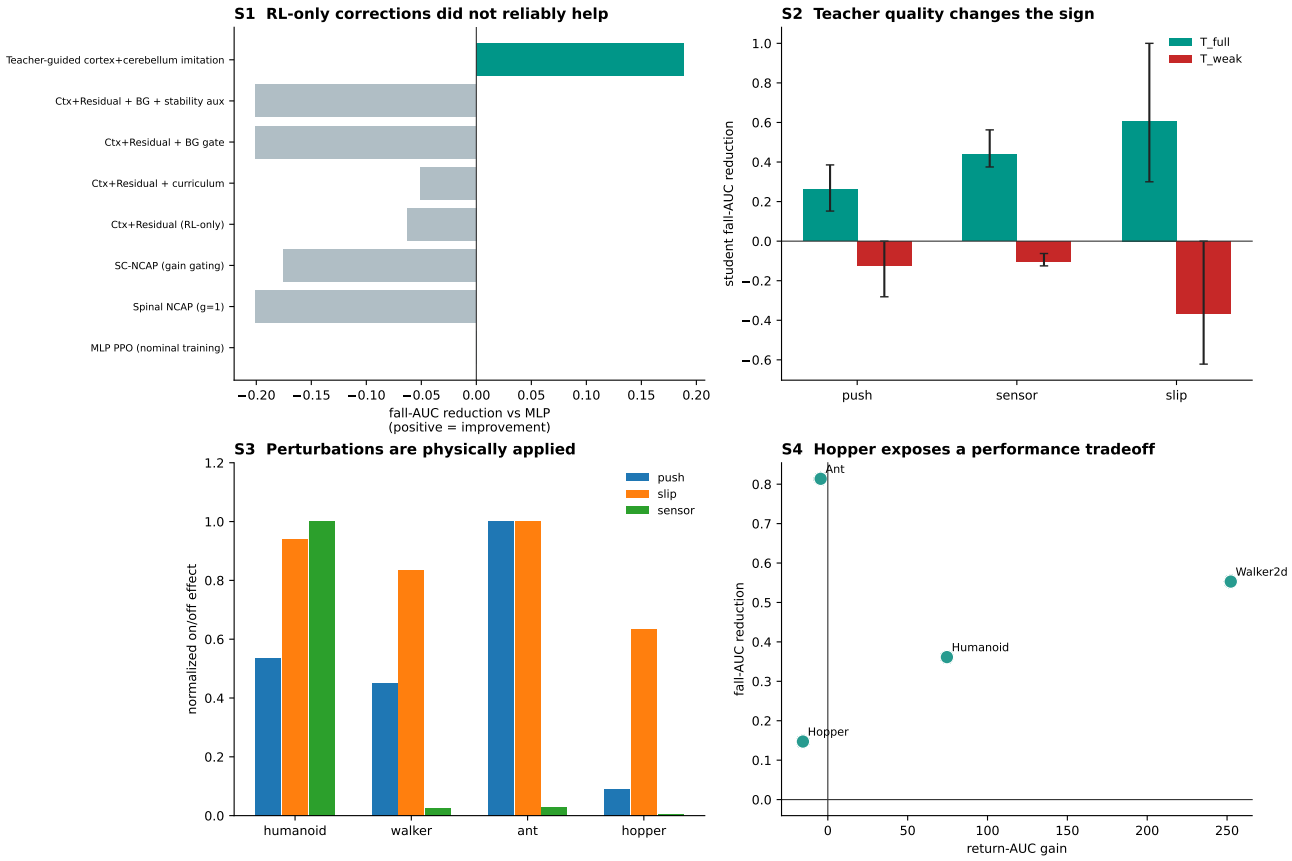

Figure 2: Supplementary checks: RL-only failure ladder, teacher-quality threshold, perturbation sanity, and cross-environment tradeoffs. In the failure-ladder panel, positive fall-AUC reduction values indicate improvement over the MLP baseline, while negative values indicate worse fall robustness.

13 Where the code lives

Core implementation files:

- `paper/experiments/sc_ncap/circuits/cerebellum.py`: bounded residual module.
- `paper/experiments/sc_ncap/policies/cortex_residual_policy.py`: cortex plus residual action composition.
- `paper/experiments/sc_ncap/scripts/train_cerebellum_imitation.py`: supervised residual imitation.
- `paper/experiments/sc_ncap/scripts/train_consolidated_cortex.py`: structural consolidation.

Acknowledgments

We thank Raymond Chua for senior mentorship, scientific guidance, and feedback on motor-control framing. We also acknowledge the BrainCAD/NCAP software infrastructure used for circuit templates, MuJoCo wrappers, evaluation, and video export.

References

[1] John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal policy optimization algorithms, 2017. URL https://arxiv.org/abs/1707.06347.